

MODELLING OF PROPELLANT SLOSH DYNAMICS AND ITS APPLICATION IN CONTROL STABILITY STUDY OF LUNAR LANDING MISSION

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Propellant sloshing during liquid-engine burn operation of a spacecraft interacts with its dynamics and influences overall stability of the spacecraft. In this paper, a high-fidelity slosh dynamics model is developed based on analogous mechanical (pendulum) model, which can be applied to study and analyze the sloshing phenomena during powered-descent mode of a lunar landing mission. The center-of-percussion study augmented by stability analysis based on the proposed slosh model as well as rate-sensitivity study are sufficient to identify the zone of predominant slosh behavior expected during the mission. As a case study, the proposed slosh dynamics model and associated stability analysis has been applied to a typical lunar landing mission.

INTRODUCTION

Sloshing phenomena associated with liquid motion in a partially filled propellant tank of a spacecraft influences its dynamics during liquid engine burn (thrusting phase) and may lead to control instability (see Reference 1). This in turn can cause violation of pointing accuracy and rate specification, specifically in landing missions, where longitudinal acceleration is quite high during powered descent phase. Therefore, it is very crucial to develop a high-fidelity slosh dynamics model as well as a proper methodology for assessing control stability in order to envisage possible interaction of propellant slosh dynamics with spacecraft attitude control system. During design phase of a mission, the more accurate the prediction of control-slosh interaction, the more beneficial will be the realization of appropriate strategy to alleviate the adverse effect of propellant sloshing on the spacecraft.

The modelling of propellant sloshing motion as well as its influence on control performance has been extensively treated in various literatures (Reference 2). One approach of modelling propellant slosh dynamics is to use computational fluid dynamics (CFD). The impeccable performance of CFD analysis in closely replicating the fluid motion inside a tank, has been brought out by various studies, which focus on evaluation of slosh force as well as different slosh parameters (see Reference 3). In fact, CFD has been applied to Guidance-Navigation and Control (GNC) simulation loop in presence of propellant sloshing (Reference 4) as well. However, the CFD based methods are computationally

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intensive, hence less attractive for assessment of control stability as well as close-loop simulation. For the purpose of control studies, dynamically equivalent mechanical model of slosh (either pendulum model or mass-spring model) is widely considered. The linearized pendulum model for slosh has been developed in References 1,5 and 6, for launch vehicle with gimballed engine configuration. In these references, the single axis spacecraft dynamics instead of three-axis coupled dynamics has been considered for the purpose of slosh study. In References 7 and 8, although independent source (reaction control system thrusters) of lateral force has been considered in addition to lateral thrust by main liquid engine, the dynamics modelling considers only single tank configuration. Reference 9 incorporates multiple tank configuration in dynamics, but the impact of lateral force in sloshing is not explicit. In Reference 10, the fundamental characteristics of various forces on spacecraft are used to develop the analogous model of sloshing, for a Cubesat mission.

The equivalent mechanical model for propellant sloshing, has been utilized to evaluate the control stability of various liquid engine-based missions. Propellant slosh modelling and control stability analysis of Solar Dynamics Observatory, has been discussed in Reference 11. In Reference 1, the stability of a thrust-vector controlled launch vehicle containing single tank is analyzed in detail and possible solution under instability has been worked out. The coupled slosh-rigid body dynamics during atmospheric flight phase has been considered in Reference 12. In this literature an analytical expression for the neutral stability is developed and also the vehicle instability has been addressed in terms of reduced-order linearized models and key transfer functions. Reference 13 derives the stability boundary of a space vehicle for various combinations like single centerline tank as well as two tank (tandem centerline, off-centerline) configuration, spacecraft with outboard reaction jets followed by effect of aerodynamic and axial acceleration on stability. But this reference is more directed towards center of percussion-based analysis. A method of limit cycle magnitude prediction followed by control-slosh stability analyses using non-linear viscous damping has been addressed in Reference 14. Model reference artificial immune system (MRAIS) based adaptive controller has been designed and developed in Reference 15, for attitude control of a spacecraft during orbital maneuvers when substantial amount of propellant from individual tanks are expected to participate in sloshing. The proposed control law has been compared with baseline optimal control system with regard to robustness towards system instabilities, failure cases as well as aggressive maneuver demands.

In this paper, we propose a novel model for propellant slosh dynamics, which is not restricted to any particular mission configuration. This model, not only considers the three-axis coupled non-linear dynamics of spacecraft and multi propellant tank configuration, but also applies to any general configuration of liquid engine (pulse /continuous firing mode, gimballed /non-gimballed structure). Moreover, a systematic approach (which consists of a set of analyses) has been developed to apprehend the slosh stability nature during various mission phases. Finally, the proposed slosh model along with the spacecraft stability analysis have been applied to a typical lunar landing mission (high acceleration during powered descent phase). Based on the results obtained, appropriate mitigation plans have been suggested for future missions having acceleration level similar to the case study under consideration.

MATHEMATICAL MODELLING OF PORPELLANT SLOSH DYNAMICS

The formulation of slosh model consists of two segments namely, analogous pendulum dynamics (representing slosh phenomena) and disturbance torque emulating the effect of sloshing on spacecraft. The dynamic nature of propellant participating in slosh is captured in terms of time varying

analogous pendulum parameters. The pendulum dynamics is formulated based on time derivative in Newtonian frame of reference i.e. inertial frame of reference. Since the propellant sloshing happens in spacecraft body frame, which is itself in motion i.e. rotating with an angular velocity ' ω ', relative to an inertial frame, therefore the slosh dynamics equation includes terms like centrifugal acceleration as well as Coriolis acceleration.

Slosh Dynamics Formulation

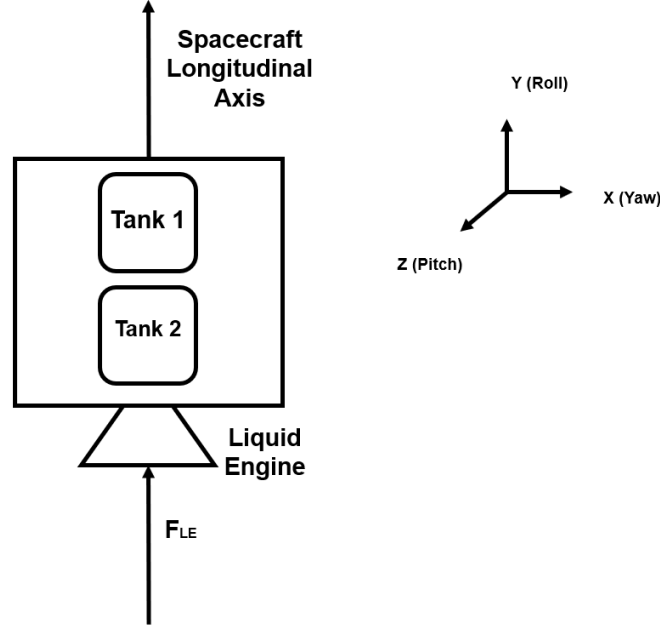


Figure 1: Configuration of Propellant Tanks in Spacecraft

To compute the forces and torques by the sloshing liquid, pendulum motion is considered in two perpendicular planes (roll-pitch and roll-yaw planes in case of spacecraft with longitudinal axis as roll axis, along which the tanks are placed) (See Figure 1, 2, 3 and 4). In this study, we have considered two sources of thrust in spacecraft, one from the liquid engine (for ΔV operation) and another from RCS thruster assembly. But the proposed slosh model is applicable to any kind of thrusting configuration (which encompasses gimballed engine case also).

For the purpose of modelling and analyses, initially a single propellant tank with only one dominant slosh mode has been considered. Further this model has been generalized for a spacecraft with several tanks with multiple slosh modes in each tank. Now, let $H(x_H, y_H, z_H)$ be the hinge point location of the pendulum *w.r.t* spacecraft center of gravity, m be the mass of the pendulum which is equal to the mass of the sloshing liquid and l be the length of the pendulum.

Positive torque about the z -axis (pitch axis) will make the slosh pendulum move towards the positive x -axis (positive yaw axis) in the XY (yaw-roll) plane, whereas, positive torque about the x -axis (yaw axis) will make the slosh pendulum move towards the negative pitch axis in the YZ (roll-pitch) plane. Let the location of the pendulum with respect to hinge along the x -axis be x_p , along y -axis be y_p and along z -axis be z_p , given as below,

Bob location w.r.t cg,
 $\vec{r} = \langle r_x \ r_y \ r_z \rangle = \langle x \ y \ z \rangle$

Bob location w.r.t hinge,
 $\vec{r}_p = \langle x_p \ y_p \ z_p \rangle$

Hinge w.r.t c.g.,
 $\vec{r}_H = \langle x_H \ y_H \ z_H \rangle$

where,

l = length of the pendulum

m = pendulum bob mass

H = hinge point

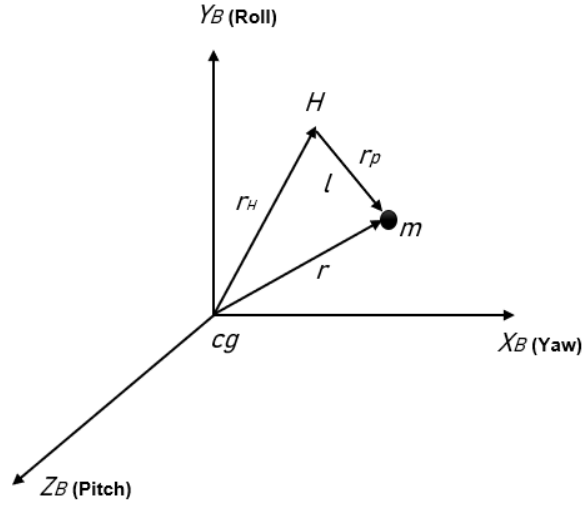


Figure 2: Spacecraft Body Coordinate System

$$x_p \approx l\psi_p \quad (1)$$

$$y_p \approx -l \quad (2)$$

$$z_p \approx -l\theta_p \quad (3)$$

where, l is the pendulum length.

Let $H(x_H, y_H, z_H)$ be hinge point coordinate of the pendulum mass, m w.r.t. center of gravity of spacecraft. Also consider, $\vec{r}$, be the location* of the pendulum w.r.t. spacecraft center of gravity which is equal to the sum of the location of the pendulum w.r.t. hinge point and the location of hinge point H , that is,

$$\vec{r} = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix} = \begin{bmatrix} x_H \\ y_H \\ z_H \end{bmatrix} + \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} = \begin{bmatrix} x_H + l\psi_p \\ y_H - l \\ z_H - l\theta_p \end{bmatrix} \quad (4)$$

The second derivative of $\vec{r}$ in inertial frame can be expressed as (see Reference 9),

$$\ddot{\vec{r}}|_I = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \ddot{\vec{r}}|_b + 2\vec{\omega} \times \dot{\vec{r}}|_b + \dot{\vec{\omega}} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (5)$$

where,

$\vec{\omega} = \langle \omega_x \ \omega_y \ \omega_z \rangle$ is the spacecraft angular rate vector, $\dot{\vec{r}}|_I$ is the first derivative of $\vec{r}$ in the inertial frame, $\dot{\vec{r}}|_b$ is the first derivative of $\vec{r}$ in the spacecraft body frame and $\ddot{\vec{r}}|_b$ is the second derivative of $\vec{r}$ in the spacecraft body frame.

*Using small angle approximation for the trigonometric functions, $\sin \epsilon \approx \epsilon$ and $\cos \epsilon \approx 1$

Also,

$$\omega \times \dot{\vec{r}}|_b = \begin{bmatrix} -\omega_y(l\dot{\theta}_p) \\ \omega_z(l\dot{\psi}_p) + \omega_x(l\dot{\theta}_p) \\ -\omega_y(l\dot{\psi}_p) \end{bmatrix} \quad (6)$$

and

$$\dot{\omega} \times \vec{r} = \begin{bmatrix} \dot{\omega}_y r_z - \dot{\omega}_z r_y \\ \dot{\omega}_z r_x - \dot{\omega}_x r_z \\ \dot{\omega}_x r_y - \dot{\omega}_y r_x \end{bmatrix} \quad (7)$$

Using Eq. 4, Eq. 7 can be further simplified as following,

$$\dot{\omega} \times \vec{r} = \begin{bmatrix} \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) \\ \dot{\omega}_z(x_H + l\psi_p) - \dot{\omega}_x(z_H - l\theta_p) \\ \dot{\omega}_x(y_H - l) - \dot{\omega}_y(x_H + l\psi_p) \end{bmatrix} \quad (8)$$

Hence, Eq. 5 can be written as,

$$\begin{aligned} \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} &= \begin{bmatrix} l\ddot{\psi}_p \\ 0 \\ -l\ddot{\theta}_p \end{bmatrix} + 2 \begin{bmatrix} -\omega_y(l\dot{\theta}_p) \\ \omega_z(l\dot{\psi}_p) + \omega_x(l\dot{\theta}_p) \\ -\omega_y(l\dot{\psi}_p) \end{bmatrix} + \begin{bmatrix} \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) \\ \dot{\omega}_z(x_H + l\psi_p) - \dot{\omega}_x(z_H - l\theta_p) \\ \dot{\omega}_x(y_H - l) - \dot{\omega}_y(x_H + l\psi_p) \end{bmatrix} \\ &+ \begin{bmatrix} \omega_x\omega_y r_y + \omega_x\omega_z r_z - (\omega_y^2 + \omega_z^2)r_x \\ \omega_x\omega_y r_x + \omega_y\omega_z r_z - (\omega_x^2 + \omega_z^2)r_y \\ \omega_x\omega_z r_x + \omega_y\omega_z r_y - (\omega_x^2 + \omega_y^2)r_z \end{bmatrix} \end{aligned} \quad (9)$$

where, in terms of separate components along yaw and pitch,

$$\ddot{x} = l\ddot{\psi}_p - 2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y r_y + \omega_x\omega_z r_z - (\omega_y^2 + \omega_z^2)r_x \quad (10)$$

and,

$$\ddot{z} = -l\ddot{\theta}_p - 2\omega_y(l\dot{\psi}_p) + \dot{\omega}_x(y_H - l) - \dot{\omega}_y(x_H + l\psi_p) + \omega_x\omega_z r_x + \omega_y\omega_z r_y - (\omega_x^2 + \omega_y^2)r_z \quad (11)$$

Sloshing in Yaw-Roll (XY) Plane

Let g be the effective acceleration produced along the Y -axis (longitudinal direction), which is given by,

$$g = \frac{f_y}{M_T} = \frac{F_{LE,y} + f_{thr,y}}{M_T} \quad (12)$$

where, f_y is the net force on spacecraft along y axis, F_{LE} is thrust due to main liquid engine, f_{thr} is the force due to reaction control system (RCS) thruster and M_T is the total mass of the spacecraft.

Let, a_x be the acceleration produced along X -axis (lateral direction) which is given by,

$$a_x = \frac{f_x + mg \sin \psi_p}{M_0} \approx \frac{f_x + mg \psi_p}{M_0} \quad (13)$$

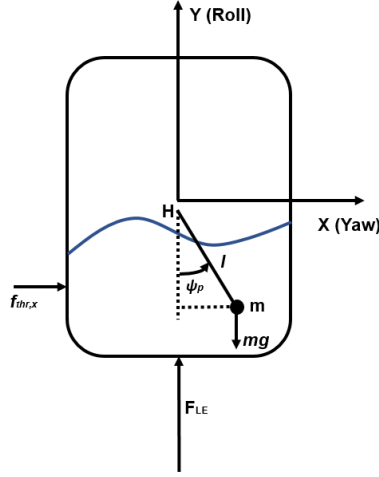


Figure 3: Sloshing in XY-plane

where, f_x is the lateral force (along the X -axis or yaw direction), f_y is the longitudinal force (along the Y -axis or roll direction) on the spacecraft due to thruster and liquid engine (LE), M_0 is the spacecraft reduced mass and ψ_p be the pendulum angle in roll-yaw plane. Therefore, the pendulum equations of motion in the xy -plane are obtained as,

$$\begin{aligned} m(a_x + \ddot{x}) &= -mg \sin \psi_p \\ \ddot{x} &= -a_x - g \sin \psi_p \end{aligned}$$

using the small angle approximation,

$$\ddot{x} \approx -a_x - g\psi_p \quad (14)$$

Substituting the value of a_x from Eq. 13 to Eq. 14,

$$\begin{aligned} \ddot{x} &= -\frac{f_x + mg\psi_p}{M_0} - g\psi_p \\ &= -g\psi_p \left(1 + \frac{m}{M_0}\right) - \frac{f_x}{M_0} \\ &= -\frac{g}{l} \left(1 + \frac{m}{M_0}\right) x - \frac{f_x}{M_0} \end{aligned} \quad (15)$$

or,

$$\ddot{x} + \frac{g}{l} \left(1 + \frac{m}{M_0}\right) x = -\frac{f_x}{M_0} \quad (16)$$

Defining equivalent sloshing frequency as,

$$\omega_{sl} = \sqrt{\frac{g}{l} \left(1 + \frac{m}{M_0}\right)} \quad (17)$$

Eq. 16 can be re-written as,

$$\ddot{x} + \omega_{sl}^2 x = -\frac{f_x}{M_0} \quad (18)$$

Introducing propellant viscous damping factor ζ in the above equation,

$$\ddot{x} + 2\zeta\omega_{sl}\dot{x} + \omega_{sl}^2 x = -\frac{f_x}{M_0} \quad (19)$$

Substituting Eq. 10 in Eq. 19,

$$l\ddot{\psi}_p - 2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y r_y + \omega_x\omega_z r_z - (\omega_y^2 + \omega_z^2)r_x + 2\zeta\omega_{sl}\dot{x} + \omega_{sl}^2 x + \frac{f_x}{M_0} = 0 \quad (20)$$

which can be solved for $\ddot{\psi}_p$ to give,

$$\ddot{\psi}_p = -\frac{1}{l} \left[-2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y r_y + \omega_x\omega_z r_z - (\omega_y^2 + \omega_z^2)r_x + 2\zeta\omega_{sl}\dot{x} + \omega_{sl}^2 x + \frac{f_x}{M_0} \right] \quad (21)$$

Substitution of (r_x, r_y, r_z) from Eq.4 in Eq. 21 implies,

$$\ddot{\psi}_p = -\frac{1}{l} \left[-2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y(y_H - l) + \omega_x\omega_z(z_H - l\theta_p) - (\omega_y^2 + \omega_z^2)(x_H + l\psi_p) + 2\zeta\omega_{sl}\dot{x} + \omega_{sl}^2 x + \frac{f_x}{M_0} \right] \quad (22)$$

The above equation can be further simplified as,

$$\ddot{\psi}_p = -\frac{1}{l} \left[-2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y(y_H - l) + \omega_x\omega_z(z_H - l\theta_p) - (\omega_y^2 + \omega_z^2)(x_H + l\psi_p) + 2\zeta\sqrt{\frac{g}{l}\left(1 + \frac{m}{M_0}\right)}l\dot{\psi}_p + g\psi_p + \frac{f_x + mg\psi_p}{M_0} \right] \quad (23)$$

As $m \ll M_0$, the above equation can be re-written as,

$$\ddot{\psi}_p = -\frac{1}{l} \left[-2\omega_y(l\dot{\theta}_p) + \dot{\omega}_y(z_H - l\theta_p) - \dot{\omega}_z(y_H - l) + \omega_x\omega_y(y_H - l) + \omega_x\omega_z(z_H - l\theta_p) - (\omega_y^2 + \omega_z^2)(x_H + l\psi_p) + 2\zeta\sqrt{\frac{g}{l}}l\dot{\psi}_p + g\psi_p + \frac{f_x + mg\psi_p}{M_0} \right] \quad (24)$$

Sloshing in Roll-Pitch (YZ) Plane

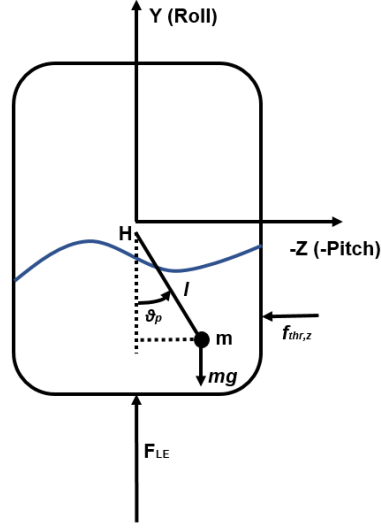


Figure 4: Sloshing in YZ-plane

Let a_z be the lateral acceleration produced along pitch direction (Z axis) given by,

$$a_z = \frac{f_z - mg \sin \theta_p}{M_0} \approx \frac{f_z - mg \theta_p}{M_0} \quad (25)$$

where, f_z be the lateral force (along the Z -axis) on the spacecraft due to thruster and liquid engine (LE) and θ_p is the pendulum angle in roll-pitch plane. The pendulum equation of motion in the yz -plane are obtained as,

$$\begin{aligned} m(-a_z + \ddot{z}') &= -mg \sin \theta_p \\ \ddot{z}' &= a_z - g \sin \theta_p \\ \text{hence, } \ddot{z} &= -a_z + g \sin \theta_p \end{aligned} \quad (26)$$

where, $z' = -z$. Using the small angle approximation above equation can be re-written as,

$$\ddot{z} \approx -a_z + g \theta_p \quad (27)$$

Substituting the value of a_z from Eq. 25 to Eq. 27,

$$\begin{aligned} \ddot{z} &= -\frac{f_z}{M_0} + g \theta_p \left(1 + \frac{m}{M_0}\right) \\ \ddot{z} - g \theta_p \left(1 + \frac{m}{M_0}\right) &= -\frac{f_z}{M_0} \end{aligned} \quad (28)$$

or,

$$\ddot{z} + \frac{g}{l} \left(1 + \frac{m}{M_0}\right) z = -\frac{f_z}{M_0} \quad (29)$$

Substituting Eq. 11 in Eq. 29 and introducing damping as in previous section, after rearranging and using $m \ll M_0$,

$$\ddot{\theta}_p = \frac{1}{l} \left[-2\omega_y(l\dot{\psi}_p) + \dot{\omega}_x(y_H - l) - \dot{\omega}_y(x_H + l\psi_p) + \omega_x\omega_z(x_H + l\psi_p) + \omega_z\omega_y(y_H - l) \right. \\ \left. - (\omega_x^2 + \omega_y^2)(z_H - l\theta) - 2\zeta\sqrt{\frac{g}{l}}l\dot{\theta}_p - g\theta_p + \left(\frac{f_z - m\theta_pg}{M_0} \right) \right] \quad (30)$$

Slosh Torque Formulation

Slosh torque $\vec{\tau}_{sl}$, is defined as,

$$\vec{\tau}_{sl} = \vec{r} \times \vec{F} \quad (31)$$

with

$$\vec{r} = \begin{bmatrix} x_H + l\psi_p \\ y_H - l \\ z_H - l\theta_p \end{bmatrix} \quad (32)$$

and

$$\vec{F} = \begin{bmatrix} mg\psi_p \\ -mg \\ -mg\theta_p \end{bmatrix} \quad (33)$$

Hence, the slosh torque components for two tanks along the x or yaw, y or roll and z or pitch axes are obtained as following,

$$\tau_{sl} = \begin{bmatrix} mg(-\theta_p y_H + z_H) \\ mg(\psi_p z_H + \theta_p x_H) \\ -mg(x_H + \psi_p y_H) \end{bmatrix} \quad (34)$$

Pendulum Dynamics and Slosh Torque Equations for General Spacecraft Configuration

Eq. 24 and Eq. 30 are derived for a single tank with single slosh mode. These equations can be generalized for the general spacecraft configuration with ‘ n_0 ’ number of slosh modes, as below:

$$\ddot{\psi}_{p_i} = -\frac{1}{l_i} \left[-2\omega_y(l_i\dot{\theta}_{p_i}) + \dot{\omega}_y(z_{H_i} - l_i\theta_{p_i}) - \dot{\omega}_z(y_{H_i} - l_i) + \omega_x\omega_y(y_{H_i} - l_i) + \omega_x\omega_z(z_{H_i} \right. \\ \left. - l_i\theta_{p_i}) - (\omega_y^2 + \omega_z^2)(x_{H_i} + l_i\psi_{p_i}) + 2\zeta_i\sqrt{\frac{g}{l_i}}l_i\dot{\psi}_{p_i} + g\psi_{p_i} + \frac{f_x + \sum_{j=1}^{n_0} m_j\psi_{p_j}g}{M_0} \right] \quad (35)$$

$$\ddot{\theta}_{p_i} = \frac{1}{l_i} \left[-2\omega_y(l_i\dot{\psi}_{p_i}) + \dot{\omega}_x(y_{H_i} - l_i) - \dot{\omega}_y(x_{H_i} + l_i\psi_{p_i}) + \omega_x\omega_z(x_{H_i} + l_i\psi_{p_i}) + \omega_z\omega_y(y_{H_i} \right. \\ \left. - l_i) - (\omega_x^2 + \omega_y^2)(z_{H_i} - l_i\theta_{p_i}) - 2\zeta_i\sqrt{\frac{g}{l_i}}l_i\dot{\theta}_{p_i} - g\theta_{p_i} + \frac{f_z - \sum_{j=1}^{n_0} m_j\theta_{p_j}g}{M_0} \right] \quad (36)$$

where, $i = 1 \dots n_0$

Similarly, Eq. 34 represents the slosh torque on the spacecraft due to single (dominant) slosh mode. For ' n_0 ' number of total slosh modes (considering all propellant tanks), the torque on the spacecraft due to sloshing can be written as,

$$\tau_{sl} = g \begin{bmatrix} \sum_{i=1}^{n_0} (m_i(-\theta_{p_i} y_{Hi} + z_{Hi})) \\ \sum_{i=1}^{n_0} (m_i(\psi_{p_i} z_{Hi} + \theta_{p_i} x_{Hi})) \\ - \sum_{i=1}^{n_0} (m_i(x_{Hi} + \psi_{p_i} y_{Hi})) \end{bmatrix} \quad (37)$$

This slosh torque is added in the total disturbance torque in the spacecraft dynamics equation.

CONTROL STABILITY ASSESSMENT

In the present work, the impact analysis of propellant sloshing on spacecraft stability is approached in multiple steps. The goal of this approach is to understand the nature of various slosh modes and their effect on spacecraft performance under various scenarios (nominal as well as perturbation cases). In case slosh instability is found to be present, the knowledge of its magnitude and duration as well as its influence on spacecraft body rate, is very crucial for deciding the mitigation strategy to contain slosh. The outline of this extensive analysis is summarized as below.

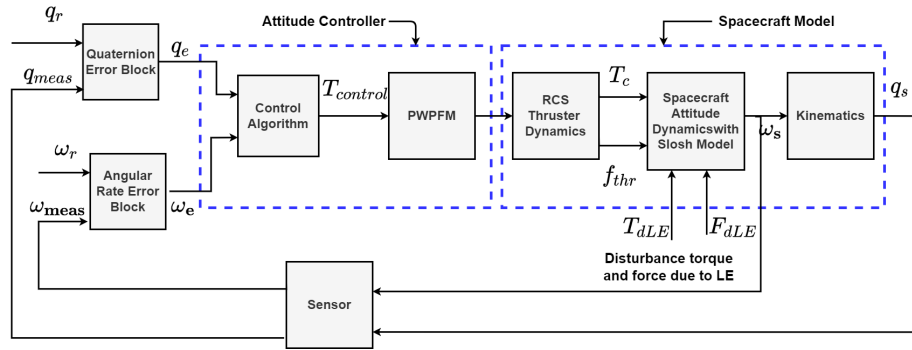


Figure 5: Block Diagram of Attitude Control Loop of Spacecraft with Reaction Control System (RCS) Thruster as Actuator

Center of Percussion Study

Center of percussion is a point forward of spacecraft center of gravity, which is a significant parameter indicating the overall slosh instability based on the location of the slosh pendulum mass as well as hinge location along the longitudinal dimension of the spacecraft. The location of center of percussion w.r.t. center of gravity gets mapped to the location of slosh pole against slosh zero, in system pole-zero configuration plot. For a given propellant tank, if the hinge location is below both center of percussion and center of gravity, then as indicated by this study, propellant tank is slosh stable (Reference 1). Similar way, stability is assured when slosh hinge as well as slosh mass are located above center of percussion. The mathematical expression of center of percussion (x_{cp}) is given by,

$$x_{cp} = \frac{I_{tr}}{M_0 l_c} \quad (38)$$

where, I_{tr} is the spacecraft inertia about any transverse axis, l_c is the distance between spacecraft center of gravity and thrust acting plane of the spacecraft.

In a nutshell, the slosh instability is plausible if the slosh mass is located between center of percussion and center of gravity. But this preliminary level study is not sufficient to confirm the stability nature of spacecraft during sloshing.

Rate Sensitivity Study

Rate sensitivity parameter governs the impact of slosh wave amplitude (pendulum angle magnitude) on spacecraft rate at any point of time. Higher the sensitivity, more is the impact of slosh oscillation on spacecraft body rate. The rate sensitivity ($\mu_{P_{rate}}$) can be formulated from the definition of acceleration sensitivity ($\mu_{P_{accn}}$), which basically highlights the effect of propellant sloshing on overall acceleration of the spacecraft. The acceleration sensitivity for i^{th} slosh pendulum can be defined as,

$$\mu_{P_{accn},i} = \frac{m_i y_{Hi} g}{I_{tr}} \quad (39)$$

and the corresponding rate sensitivity expression is given by,

$$\mu_{P_{rate},i} = \frac{\mu_{P_{accn},i}}{\omega_{sl,i}} \quad (40)$$

where, $\omega_{sl,i}$ is the slosh frequency of i^{th} pendulum.

Short Period Analysis

- Derivation of States Space Model:

In this exercise, the overall spacecraft attitude dynamics including propellant slosh model (Eq. 35 - Eq. 37) is linearized to obtain the state space model of the system (spacecraft rigid body dynamics is based on *Euler moment equation*, Reference 9). Here various slosh parameters as well as the physical parameters of the spacecraft are considered to be constant for a given time interval. Thus, time frozen dynamics is obtained in matrix form for various time intervals of the mission. Depending upon the time of interest, appropriate state space model is chosen for further stability analysis. The states space model of the spacecraft attitude dynamics in presence of propellant sloshing is formulated as below:

$$\begin{aligned} \dot{X} &= AX + BU \\ Y &= CX + DU \end{aligned} \quad (41)$$

The state vector $X \in \mathbb{R}^N$ is defined as below,

$$X = [\psi \ \phi \ \theta \ \omega_x \ \omega_y \ \omega_z \ \psi_{p_i} \ \theta_{p_i} \ \dot{\psi}_{p_i} \ \dot{\theta}_{p_i}]' \quad (42)$$

where, $N = 6 + 4 * n_0$ and subscript ' i ' denotes the i^{th} pendulum. The input vector $U \in \mathbb{R}^6$ is given by,

$$U = [T_c \ f_{thr}]' \quad (43)$$

T_c is the control torque due to RCS thruster(s) acting on the spacecraft.

$Y \in \mathbb{R}^N$ denotes output of the state space model, $C \in \mathbb{R}^{N \times N}$ is the observation matrix and

$D \in \mathbb{R}^{N \times 6}$ is a null matrix.

For the matrix $A \in \mathbb{R}^{N \times N}$,

$$\begin{aligned}
A_{ij} &= I_{3 \times 3}, \quad i = 1 \dots 3 \text{ \& } j = 4 \dots 6 \\
A_{ij} &= -\mu_{Paccn, i-x}, \quad i = 4 \text{ \& } j = 7 + n_0 \dots 6 + 2n_0 \\
A_{ij} &= -\mu_{Paccn, i-z}, \quad i = 6 \text{ \& } j = 7 \dots 6 + n_0 \\
A_{ij} &= I_{2n_0 \times 2n_0}, \quad i = 7 \dots 6 + 2n_0 \text{ \& } j = 7 + 2n_0 \dots 6 + 4n_0 \\
A_{ij} &= \text{diag}\{-2\zeta_1 \omega_{sl,1}; \dots; -2\zeta_k \omega_{sl,k}; -2\zeta_1 \omega_{sl,1}; \dots; -2\zeta_k \omega_{sl,k}\}, \\
&\quad i, j = 7 + 2n_0 \dots 6 + 4n_0 \text{ \& } k = 1 \dots n_0 \\
A_{ij} &= A_{\psi_p}, \quad i = 7 + 2n_0 \dots 6 + 3n_0 \text{ \& } j = 7 \dots 6 + n_0 \\
A_{ij} &= A_{\theta_p}, \quad i = 7 + 3n_0 \dots 6 + 4n_0 \text{ \& } j = 7 + n_0 \dots 6 + 2n_0
\end{aligned} \tag{44}$$

where, $I_{p \times p}$ is identity matrix of size $p \times p$.

$\mu_{Paccn, i-x} = \frac{m_i y_{Hi} g}{I_x}$, corresponds the acceleration sensitivity in ‘x’ (yaw) axis, I_x being the yaw moment of inertia of the spacecraft.

Similarly, $\mu_{Paccn, i-z} = \frac{m_i y_{Hi} g}{I_z}$, corresponds the acceleration sensitivity in ‘z’ (pitch) axis, I_z being the pitch moment of inertia of the spacecraft.

The terms A_{ψ_p} and A_{θ_p} as following:

$$\begin{aligned}
(A_{\psi_p})_{kk} &= \frac{1}{l_k} \left(-(1 + \frac{m_k}{M_0})g - (y_{Hk} - l_k) \mu_{Paccn, k-z} \right), \quad k = 1 \dots n_0 \\
(A_{\psi_p})_{kq} &= \frac{1}{l_k} \left(-\frac{m_q}{M_0}g - (y_{Hk} - l_k) \mu_{Paccn, q-z} \right), \quad k, q = 1 \dots n_0 \text{ \& } k \neq q
\end{aligned} \tag{45}$$

Likewise,

$$\begin{aligned}
(A_{\theta_p})_{kk} &= \frac{1}{l_k} \left(-(1 + \frac{m_k}{M_0})g - (y_{Hk} - l_i) \mu_{Paccn, k-x} \right), \quad k = 1 \dots n_0 \\
(A_{\theta_p})_{kq} &= \frac{1}{l_k} \left(-\frac{m_q}{M_0}g - (y_{Hk} - l_k) \mu_{Paccn, q-x} \right), \quad k, q = 1 \dots n_0 \text{ \& } k \neq q
\end{aligned} \tag{46}$$

The other terms of A matrix are zero. Matrix $B \in \mathbb{R}^{N \times 6}$ is given by,

$$\begin{aligned}
B_{ij} &= \text{diag} \left\{ \frac{1}{I_x}; \frac{1}{I_y}; \frac{1}{I_z} \right\} \quad i = 4 \dots 6 \text{ \& } j = 1 \dots 3 \\
B_{ij} &= \left[-\frac{z_{Hi}}{l_i I_y} \frac{(y_{Hi} - l_i)}{I_z l_i} - \frac{1}{M_0 l_i} \right], \quad i = 7 + 2n_0 \dots 6 + 3n_0 \text{ \& } j = 2 \dots 4 \\
B_{ij} &= \left[\frac{(y_{Hi} - l_i)}{I_x l_i} - \frac{x_{Hi}}{l_i I_y} \quad 0 \quad 0 \quad 0 \quad \frac{1}{M_0 l_i} \right], \quad i = 7 + 3n_0 \dots 6 + 4n_0 \text{ \& } j = 1 \dots 6
\end{aligned} \tag{47}$$

Entries of matrix B , other than those mentioned above are all zero.

- Stability analysis with short period dynamics (Time Slice Approach):

The spacecraft mass properties and slosh parameters for a given time instant of the mission, are chosen for stability study in frequency as well as time domain*. The *digital autopilot*

*In this paper the short period study, center of percussion study as well as rate sensitivity results are shown for yaw plane case only. Similar results exist for pitch plane as well.

(DAP) loop considered for this study, is shown in Figure 5. q_r and ω_r are reference quaternions and reference angular rate respectively while q_{meas} & ω_{meas} are measured attitude and body rate, obtained from sensors. T_{dLE} and F_{dLE} are disturbance torque and force due to liquid engine. T_C , f_{thr} are the realized output torque and force from RCS thruster(s). These RCS thrusters are in turn driven by PWPFM (pulse width pulse frequency modulator) output. This whole stability analysis is repeated at regular time intervals within the time zone of interest.

Time Domain Simulation with Non-linear dynamics

In this study, the slosh and spacecraft parameters vary every dynamic cycle. This study considers the complete non-linear dynamics model of various blocks in the control loop (refer Figure 5) as well as additional blocks like: Thruster Selection logic (TSL), and onboard delay. The experimental characterization of liquid engine as well as thrusters (in terms of rise time, fall time, chamber pressure, specific impulse, mass flow rate and propellant mixture ratio) is also considered in this study. Further, the onboard software implementation logics are appended to this ground simulation software to precisely emulate the expected response from system during the mission.

Further, the reference variables (quaternions and body rates) in such simulations are time varying and computed every major cycle based on the given orbit profile or the closed loop guidance demand, unlike short period analysis (where the control references are frozen for a given time interval). The goal of this kind of comprehensive study is to capture the end-to-end performance of the mission under nominal as well as off-nominal scenarios.

Parametric Perturbation Study

This study is about sensitivity assessment of various parameters under consideration for a particular mission. Firstly, a group of perturbation parameters are shortlisted by analyzing the impact of individual parameter perturbation on system response. Thereafter, the mission simulation is performed by perturbing those parameters within the pre-defined bounds. Extensive simulations are performed in frequency domain as well as time domain and it gives the overall impression of the system's stability and robustness under perturbation of identified parameters.

APPLICATION OF PROPOSED SLOSH MODEL AND STABILITY ANALYSIS

For the purpose of validation of slosh model as well as demonstration of proposed stability assessment methodology, the closed loop configuration of a typical lunar landing mission has been considered* and attitude control stability has been analyzed in this paper.

In this study, bi-propellant type propulsion system is considered with tanks in tandem centerline configuration (refer Figure 1), which is a quintessential propulsion configuration for various lunar landing missions. Here, the two propellant tanks, *Tank 1* (fuel) and *Tank 2* (oxidizer), are placed along the longitudinal axis of the spacecraft (here roll (Y) axis, see Figure 1). During the powered descent phase of landing, substantial propellant mass from both of the tanks start to participate in sloshing oscillation as a response to longitudinal acceleration and lateral force from the main liquid engine as well as attitude control thrusters. From the experimental data of the propellants tanks under study, it is found that the first slosh mode of each tank is the most dominant and hence considered for further study (therefore ' $n_0 = 2$ ' in Eq. 35 - Eq. 37).

*slosh parameters for the analysis, are obtained from CFD study as well as experimental setup

Center of Percussion Study

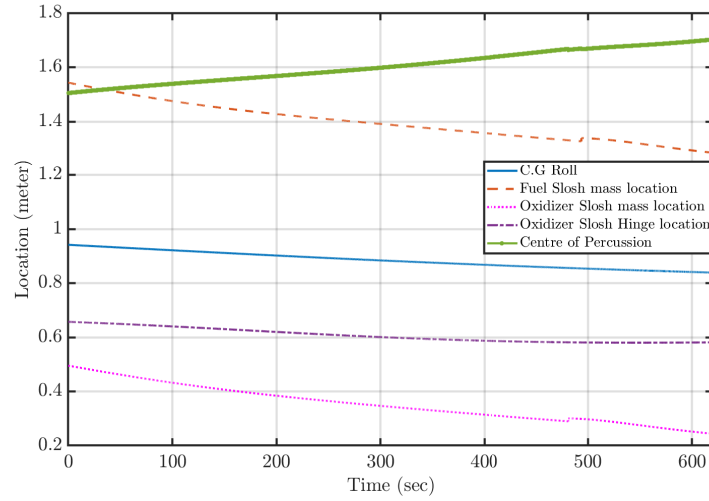


Figure 6: Center of Percussion

The center of percussion study result from Figure 6, indicates the possible existence of instability in fuel slosh mode. On the other hand, the oxidizer slosh mode is stable and hence may not contribute to slosh excitation. But the center of percussion-based study is not sufficient to conclude regarding the overall stability of lander throughout mission phases, rather it depends significantly on the dynamic condition (considerable longitudinal acceleration and lateral force) of the mission.

Rate Sensitivity Study

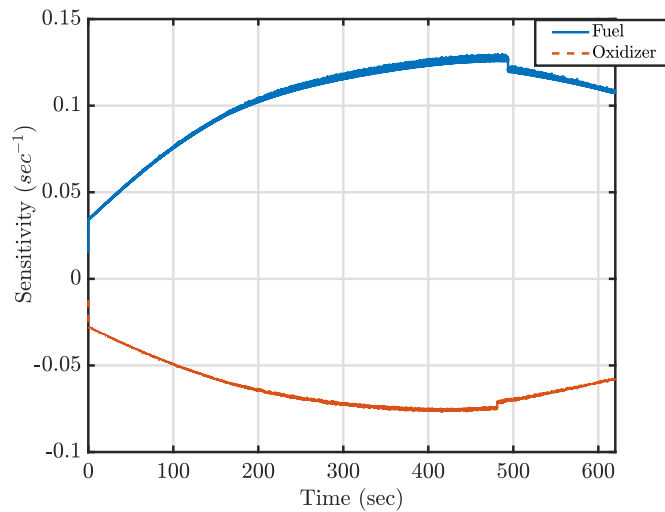


Figure 7: Rate Sensitivity

From the rate sensitivity study in Figure 7, it can be observed that the rate sensitivity is less during the beginning phase of the mission and reaches its maximum near 500 seconds. The slosh instability, if exists, will manifest itself as amplified oscillation of body rate around this time instant. This is further corroborated through short period analysis and non-linear time domain simulation.

Short Period Analysis

The short period study results for *yaw axis* are analyzed and included here. The frequency domain analysis at various time instants is shown in Figure 8 . The Nyquist diagram clearly illustrates two slosh loops, each corresponding to one of the tanks. The fuel slosh mode is near the instability point $(-1 + j0)$, as supported from center of percussion study. The stability of rigid body mode and oxidizer mode is ensured throughout the powered descent.

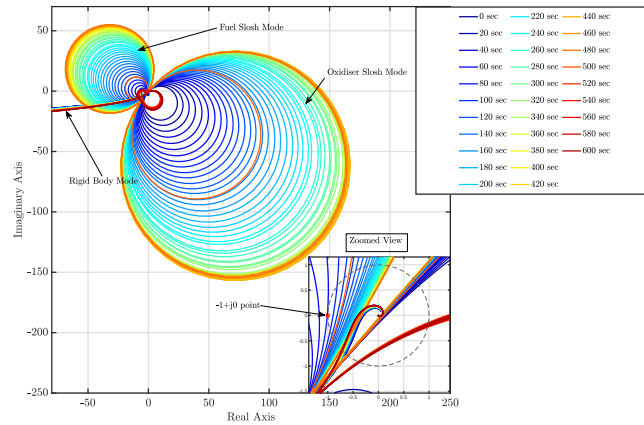


Figure 8: Nyquist Diagram During Powered Descent Phase

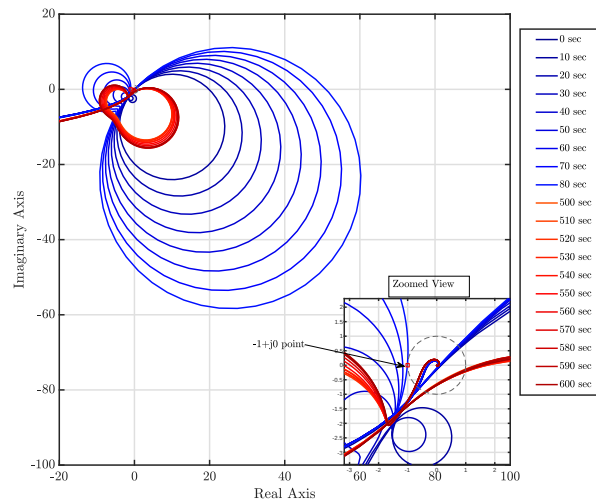
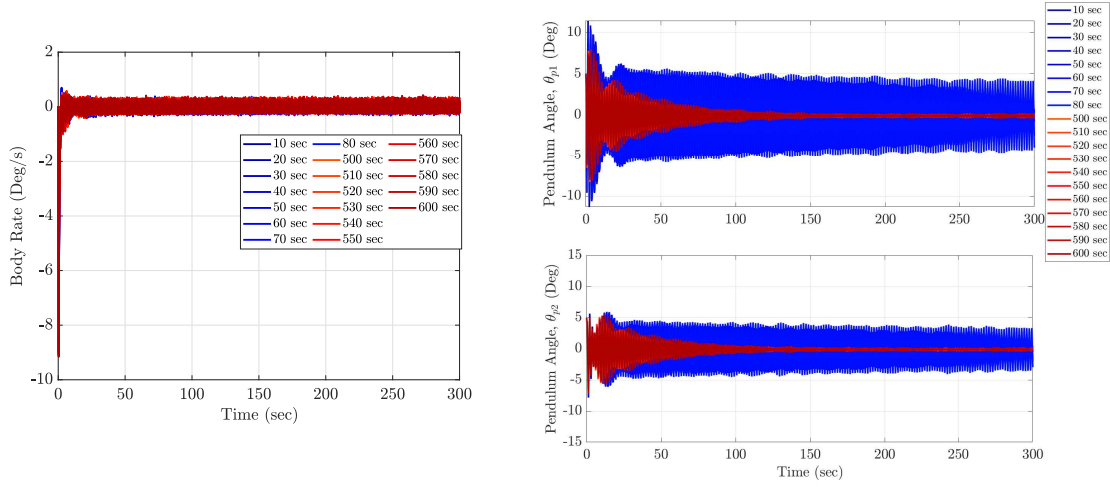
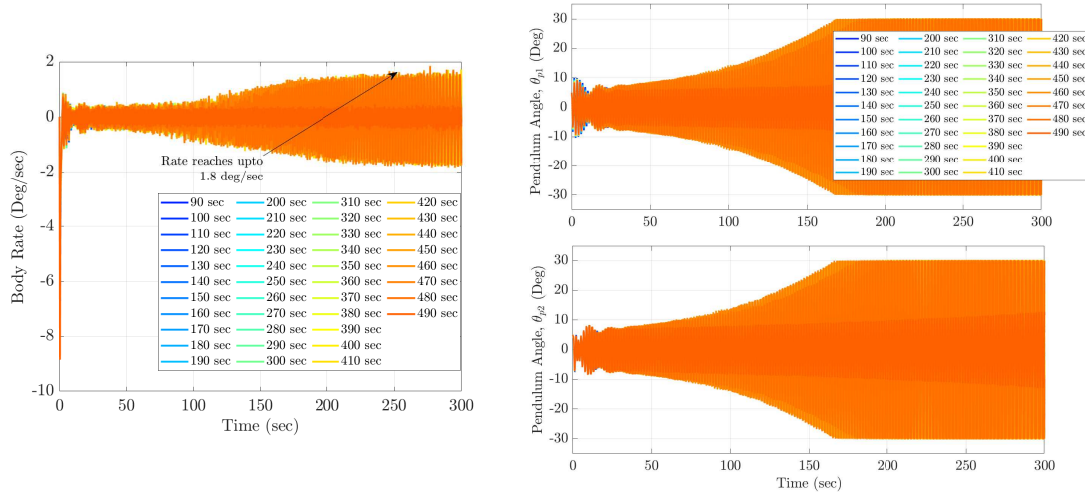


Figure 9: Slosh Stable Time Zone

Table 1: Summary of Frequency Domain Study

Time Zone	Nature of Stability	Instability of Fuel Slosh Mode (in terms of Phase Margin, deg)
Zone 1 (10 sec - 90 sec)	stable	Nil
Zone 2 (90 sec - 500 sec)	unstable	-102.48 ⁰ (Maximum) at 470 sec & corresponding slosh frequency 0.51 Hz
Zone 3 (500 sec - 620 sec)	stable	Nil

**Figure 10:** Short Period Time Domain Simulation: Slosh Stable Zone**Figure 11:** Short Period Time Domain Simulation: Slosh Unstable Zone

Frequency Domain Study Table 1 highlights the stability margin and stable/unstable zone duration as obtained from frequency domain study. After 500 seconds, the *propellant acquisition system* (PAS) provides additional damping which in turn aids to the improvement in overall stability.

Time Domain Study The short period time domain study is performed by simulating the spacecraft dynamics for a fixed duration (300 seconds here) keeping the spacecraft and slosh parameters frozen corresponding to a particular time instant of the mission. This simulation is repeated for various time instants (at regular interval) of the thrusting phase. Figure 10 shows the stable zone (zone 1 & 3) whereas Figure 11 represents the unstable zone (zone 2) due to sloshing. This study corroborates the vehicle stability analysis in frequency domain. The instability in time domain is manifested in terms of sustained oscillation of 1.8 deg/sec and pendulum angle reaching to 30° .

Time Domain Simulation with Non-Linear Dynamics

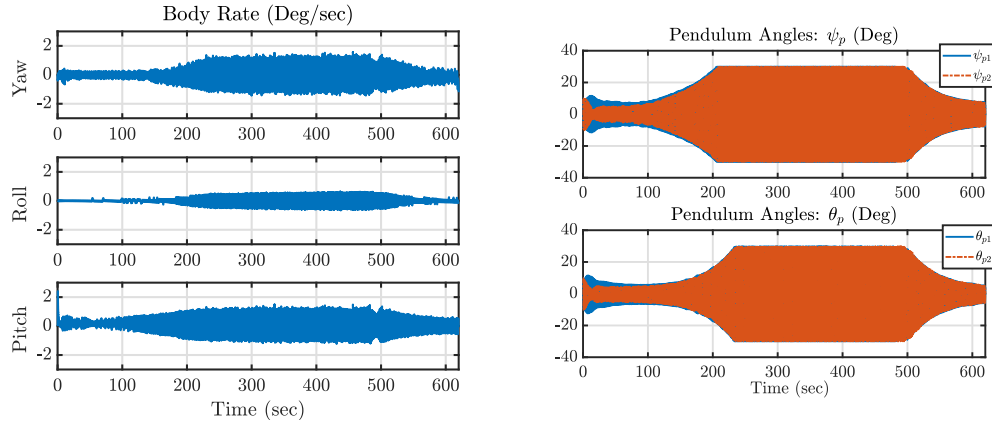


Figure 12: 6-DOF Study: Attitude Control Loop Simulation Results

The short period analyses combined with center of percussion study and rate sensitivity study identify the zone of instability for the mission. The presence of suitable mission condition, engine thrust misalignment and thrust uncertainty etc. ultimately dictate the fate of slosh triggering, which is replicated through 6-DOF simulation (Figure 12). As shown in the figure, the slosh effect starts to build near 200 seconds and continues up to 500 seconds, after which the slosh oscillation begins to damp out (see pendulum angles) under nominal scenario. This zone is clearly a subset of instability zone predicted from short period study.

Parametric Perturbation Study

The statistical inference on attitude control loop behavior, depending upon uncertainty of various parameters (slosh as well as lander parameters) can be drawn from parametric perturbation study. First, a number of slosh sensitive parameters of the spacecraft are identified, for which sensitivity study has been carried out both in frequency as well as time domain (considering single parameter perturbation at a time). Based on this study, a list of parameters is finalized, having the most significant contribution in sloshing and overall spacecraft stability. For this final set of parameters, frequency domain analyses have been performed at various time instants, perturbing those parameters in the range specified in Table 2*. The frequency domain study gives a clear indication regarding

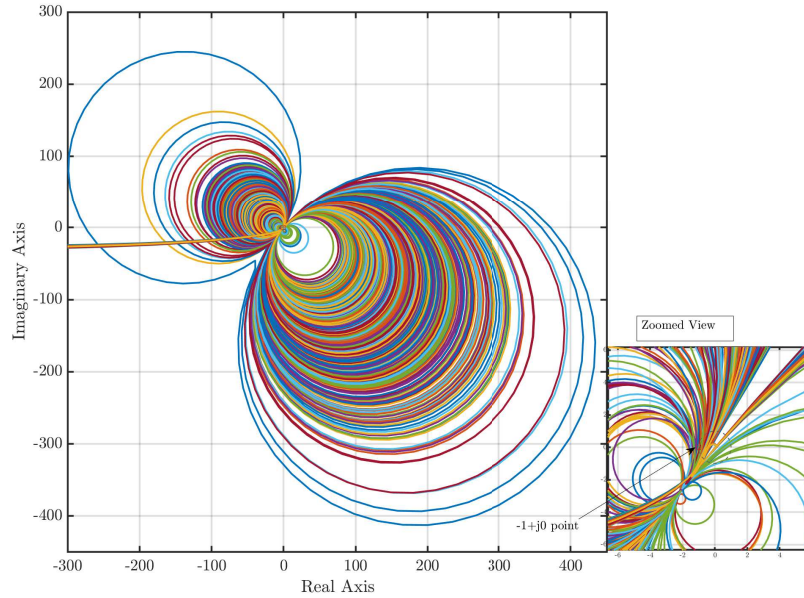
*Uncertainty in various slosh parameters are obtained based on experimental study

Table 2: Perturbation Range of Slosh Sensitive Parameters for Onboard Matching Study

Serial Number	Parameter	Corresponding Entity	Perturbation Range
1	Slosh mass	Fuel Tank Oxidizer Tank	$\pm 5\%$
2	Slosh frequency	Fuel Tank Oxidizer Tank	$\pm 5\%$
3	Viscous Damping	Fuel Tank Oxidizer Tank	-10%
4	Hinge Location from Tank Bottom	Fuel Tank Oxidizer Tank	$\pm 5\%$
5	Center of Gravity	Lander	± 10 mm (along yaw and pitch)
6	Moment of Inertia (along each body axes)	Lander	$\pm 10\%$

maximum instability, which can occur at different time-stamps. Figure 13 gives an example of instability which can possibly appear at 470 seconds from powered descent start under perturbation of specified parameters.

The slosh sensitive parameters are varied within the pre-specified bound, in time domain simulation as well. Among the results obtained from various simulation runs (refer Figure 14), a number of cases have been identified which can closely replicate the in-flight behavior.

**Figure 13:** Perturbation Study Result during Powered Descent Phase: Frequency Domain

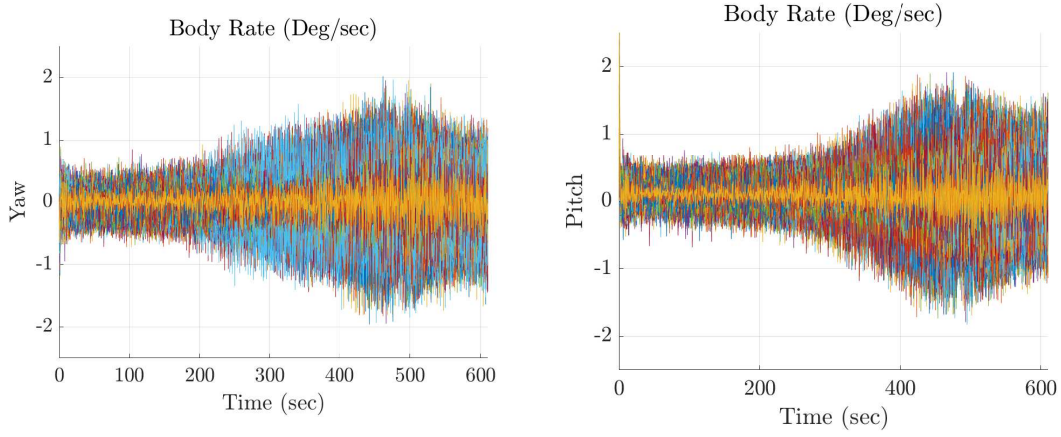


Figure 14: Perturbation Study Result during Powered Descent Phase: Time Domain

SLOSH MITIGATION PLANS FOR FUTURE MISSIONS

Following mitigations plans can be adopted in future missions to address possible interaction between propellant sloshing and spacecraft dynamics under high acceleration condition (order of 3 m/s^2 and beyond).

- **Hardware Compensation Method (Baffle Design)**

Under this method, additional damping requirement from baffle at various time instants of the mission is generated, which is further provided as input towards mechanical design of baffle. The necessary steps for the damping requirement generation can be summarized as below.

1. Calculation of allowable pendulum angle $\mathcal{T}_p$ from the following equation,

$$\mathcal{T}_p = \frac{\omega_{spec}}{\sum_{i=1}^{n_0} |\mu_{Prate,i}|} \quad (48)$$

where, ω_{spec} is the body rate specification for various phases of the mission (it is generally decided during mission design depending upon requirements for various sensor operations).

2. Evaluation of additional damping (ζ_{net}) required to stabilize the unstable slosh mode from Nyquist study
3. Finally, computation of ‘one-degree pendulum equivalent baffle damping’ (ζ_{1deg}) as below,

$$\zeta_{1deg} = \frac{\zeta_{net}}{\sqrt{\mathcal{T}_p}} \quad (49)$$

The profile of ζ_{1deg} at various time instances can be used for realization of baffle hardware.

- **Software Compensation Method**

This compensation philosophy encompasses the tuning of attitude controller gains as well as filter (lead-lag/notch) design. An end-to-end non-linear time domain simulation with the designed gains/filter, followed by critical analysis is necessary to capture the impact on rigid body performance.

From the stability margin seen in Table 1, a combination of above-mentioned compensation techniques is recommended for missions with high longitudinal acceleration demand during thrusting phase, so that minimum compromise on rigid body response can be ensured throughout the mission.

CONCLUSION

In this paper, propellant sloshing motion has been modelled using pendulum dynamics analogy. This model is generic in the sense that it can be augmented to any kind of propellant tank configuration as well as actuator assembly. Even the number of slosh modes per propellant tank, is a design input. Another contribution of this study is the development of a systematic approach to address the overall stability of a spacecraft. For validation and establishment of the proposed slosh model and stability assessment philosophy, a typical lunar landing mission has been considered as case study. The results of this case study clearly give a high level of confidence for its application in pre-flight control design and performance study of future space missions. Also, the comprehensive exercise of spacecraft stability assessment through the proposed slosh dynamics model is sufficient to replicate the onboard behavior of any mission specifically with *high-g* condition during liquid engine burn. Based on the analysis results, necessary strategy (discussed above) can be adopted to mitigate slosh instability while ensuring the desired control performance throughout the mission span.

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